SIMONE ROMITI

Machine Learning Engineer | Computational Scientist

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Machine learning engineer and computational scientist with a PhD and 4+ years of postdoctoral research experience building production-grade models and software for high-dimensional inference, optimization, and simulation. Designs, trains, and validates deep learning systems in PyTorch – including physics-informed and generative architectures – end-to-end: data pipelines, model design, GPU training infrastructure, and statistical validation. Main author and maintainer of open-source Python/C++ libraries with CI/CD and versioned releases.

Resident in Switzerland with a B permit.

EDUCATION

PhD in Physics | *Rome, Italy* | Nov. 2018 – Oct. 2021 | title awarded in April 2022

M.S. in Physics | *Rome, Italy* | Oct. 2016 – Oct. 2018 | Grade: 110/110 *cum laude* | GPA 3.98

B.S. in Physics | *Rome, Italy* | Oct. 2013 – Jul. 2016 | Grade: 110/110 *cum laude* | GPA 3.85

WORK EXPERIENCE

University of Bern, Switzerland | Apr. 2024 – present | *Postdoctoral Researcher* | ML/software developer for scientific applications

University of Bonn, Germany | Nov 2021 – Mar 2024 | *Postdoctoral Researcher* | Software developer for scientific applications

COMPETENCES

- **Machine learning, model development**: Physics-Informed Neural Networks, convolutional networks, and diffusion models in PyTorch, stochastic minimization.
- **Probabilistic inference and algorithm design** for high-dimensional Bayesian models: Markov Chain Monte Carlo, Langevin dynamics, Nested Sampling, and spectral methods for time series.
- **Statistical analysis and inference pipelines for terabyte-scale datasets**, from data production through inference: autocorrelation analysis, Akaike-weighted model averaging, and rigorous error budgeting.
- **Performance engineering** for compute-heavy pipelines via CPU/GPU parallelization, with scaling-aware optimization of memory footprint and runtime cost.
- **Technical communication**: peer-reviewed publications, invited conference talks, and workshop organization, translating complex methods for varied audiences.
- **Mentoring** and technical onboarding for Master's and PhD students, including code review and reproducible-research practices.

SOFTWARE TOOLS

Python libraries - PyTorch, NumPy, SciPy, Pandas, Matplotlib, Plotly, Streamlit, SymPy, Jupyter

Languages - Bash, C, C++, Python, R, SQL, LaTeX

DevOps & Tools - Git, GitHub Actions, CI/CD, Docker, Quarto Markdown, Agentic Coding

HPC & Systems - CUDA, MPI, OpenMP, SLURM, EasyBuild

Spoken Languages - Italian (native), English (proficient), German (basic)

ACHIEVEMENTS

- Developed a novel **Physics-Informed Neural Network** algorithm for solving the Schrödinger equation, implemented end-to-end in PyTorch.
- **Principal Investigator** for $O(10^6)$ GPU-hour compute allocations on CSCS ALPS and GCS JUPITER (2026), competitively secured through technical proposal review.
- Designed and operated **large-scale stochastic simulations** on international HPC clusters, improving sampling efficiency.
- **Reduced algorithmic complexity** of a class of high-dimensional integral estimators from exponential to $O(N \log N)$ by redesigning the method around parallelized Fast Fourier Transforms.
- Derived and implemented a **numerical method achieving machine-precision accuracy** for a class of constrained optimization problems.
- Designed a **hybrid classical-quantum algorithm** (Monte Carlo + quantum computing) for high-dimensional integral estimation.

SELECTED PROJECTS

- **lattice_data_tools** – Python toolkit for production and statistical analysis of large-scale simulation data. Sole author and maintainer; CI/CD and versioned releases.
- **adiabatic_PINNs-suN** – PyTorch-based Physics-Informed Neural Network solver for high-dimensional differential equations; reference implementation for a sole-authored publication.
- **su2** – C++ templated library for Monte Carlo simulation with OpenMP parallelization. Core maintainer.
- **tmLQCD** – Large-scale C library for GPU-accelerated simulations (CUDA + MPI). Contributor.
- **Patient management app (Streamlit)** – Designed and deployed a private web application for a nutrition practitioner to manage patient records (demographics, weight, BMI) and visit history, with search and nutritional-conversion tools across a food database. Sole developer: requirements gathering, UI design, and deployment.